An Overview over Explainable AI
Introduction

Lecture: Explainable Artificial Intelligence, ST 2026
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Outline

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Motivation: Why We Need Explanations

- What is Explained: (Machine Learning) Models
- The Problem with Black-boxes

Definitions

Course Outline

Criteria for Good Explanations

Summary

Motivation: Why We Need Explanations

SCHUFA-
BonitätsAuskunft

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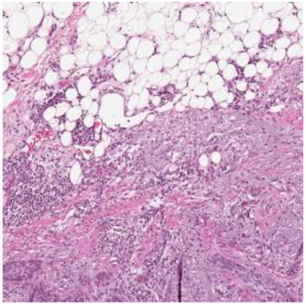


[9, Fig. 6]

Where we need explainable artificial intelligence (XAI)

Use-cases:

- End users:
  - (appropriate!) **Trust**, informed consent
  - Onboarding
  - **Recourse**
- For developers and expert users:
  - **Debugging**
  - Knowledge retrieval
- For assessors:
  - Compliance *with law and standards*
  - **Assessment**, *e.g., wrt. safety, fairness*

Application Fields:

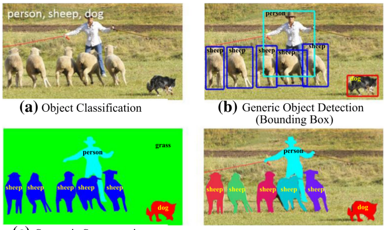
Wherever automated decisions influence human well-being!

- **Ranking** systems *(credits, applications, ...)*
- **Medical** assistant systems
- Automated **driving**
- **Military** decision systems
- **HMI** in Production
- ...

What is a Model

function

$$\overbrace{y}^{\text{Output}} = \underbrace{f_\theta}_{\text{Model (with param. }\theta)}\big(\overbrace{x}^{\text{Input feature (vector)}}\big)$$



(a) Object Classification
(b) Generic Object Detection (Bounding Box)
(c) Semantic Segmentation
(d) Object Instance Segmetation

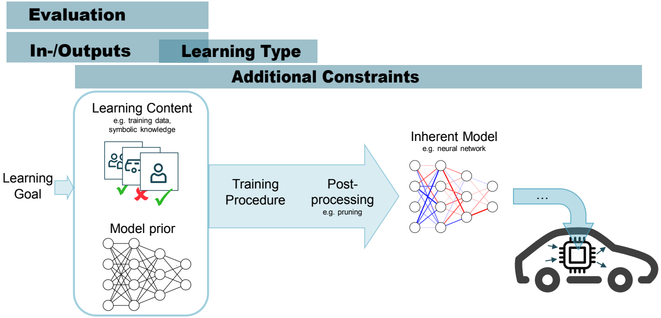
Inputs x: *images, personal data, knowledge base + query, action domain description*.

Outputs y:

- **classification** (multi-class, multi-output)
  used, e.g., for decease prediction, document/text classification,anomaly detection, action planning, next word (ChatGPT).
- **regression**
  used, e.g., for prediction of weather, demand, credit scoring (SCHUFA).
- **"multi-decision"** = generalization to, *e.g., driveable area / obstacle detection, optical character recognition (OCR)*

Scientific Modelling: The artificial intelligence (AI) System

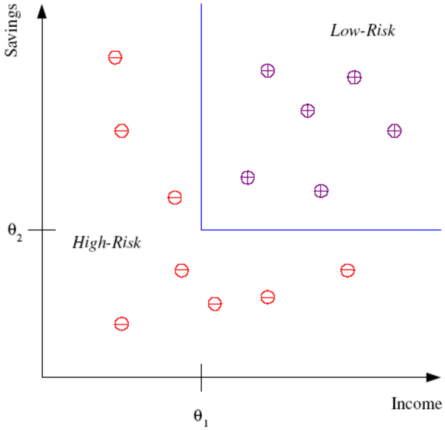
1. Observe (x, y)
2. Formulate theory
3. Build formal model f
4. Infer consequences
5. Test experimentally
6. Revise theory



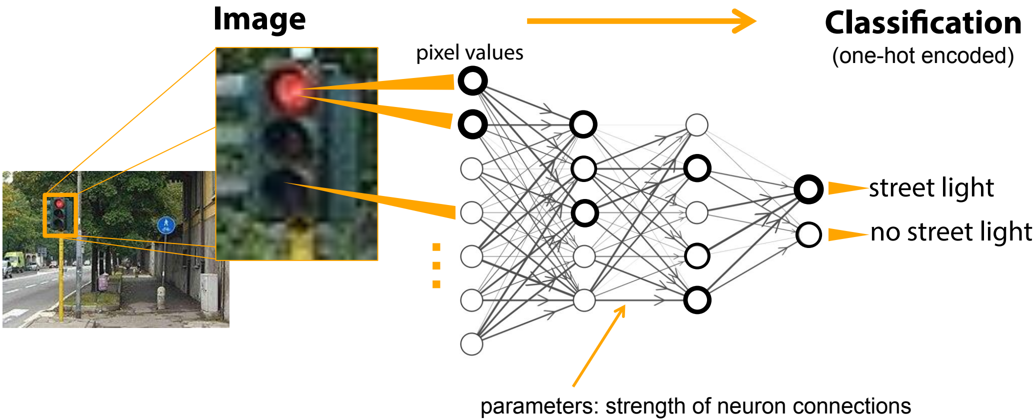
Example: Rules

Example (Credit Risk)

$$\text{HasLowRisk}(\text{income}, \text{savings}) = ((\text{income} > \theta_1) \wedge (\text{savings} > \theta_2))$$



Example: deep neural network (DNN)



Example: DNN

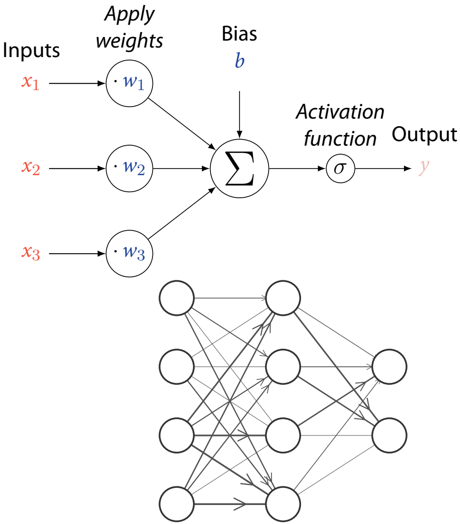
Definition (Feedforward DNN)

A DNN is

- a function $f_\theta : \mathbb{R}^n \to \mathbb{R}^m$
- built as a concatenation of neuron functions $f_{\mathbf{w},b}$

Feedforward DNN = no cycles in its directed graph of neurons.

parameters of all neurons together are the (optimizable) parameters of the DNN (θ).



The Problem with Black-boxes

Husky **image**
misclassified as *Wolf*

features most influential
for the decision

- *Overfitting*: **memorize the training data!**
- *Replicating biases*: Automatic optimization only allows to leverage **correlations** in the data.
- *Clever Hans effect* due to *shortcut* learning
  Goodhart's law [3]: "When a metric is used as a target, it ceases to be a good metric."

$\Rightarrow$ **Why** output y? **How** does the model work in general?

Outline

Important Terms in XAI

Definition (Understanding)

successful update of mental model;
can be
mechanistical = how it works, or
functional = what is its purpose.

“Good explanations must be relevant to a, potentially implicit, human question as well as relevant to the mental model of the *explainee*.” [1, 8]

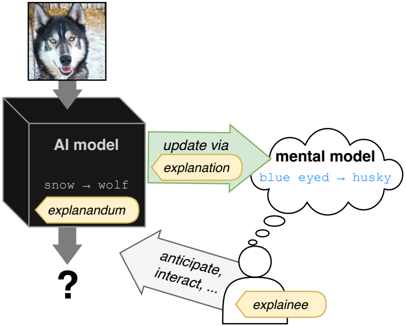
Definition (Explainable decision system)

There exists a mechanism providing an explanation (= *explanator*)
to a human (= *explainee*) allowing them to
understand one of (= *explanandum*) [6]

- the **model**’s learned information,
- the **influence** on model behavior, or
- the system’s **reasoning**.

Explainability/interpretability = degree of understanding [5].

Important Terms in XAI



Definition (Explainable decision system)

There exists a mechanism providing an explanation (= *explanator*) to a human (= *explainee*) allowing them to *understand* one of the (= *explanandum*) [6]

- **influence** on behavior (*local*),
- **model**'s learned information (*global*), or
- system's **reasoning** (*global*).

Explainability/interpretability = degree of understanding [5].

⇒ **What is the question (explanandum)?**

Excursion: Mind the Human!

how humans explain behavior (i.e., state changes) / actions:

Unintentional behavior:

- ***mere cause* explanations**:

  "My eyes feel heavy because I am tired."

Intentional behavior: explanation of

- ***reason***: beliefs & desires

  "I do X because I think Y",

  "I do X because I want Y".

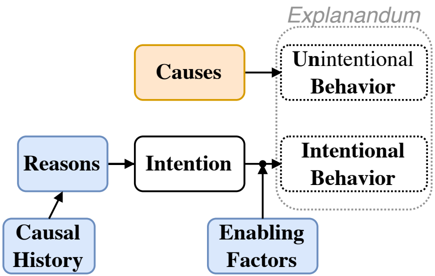
- ***causal history***: preceeding factors

  "I do X because I have trait Y",

  "I do X because I once experienced Y".

- ***enabling factor***:

  How was it possible that X was done?



Example

"Why do you listen to this lecture?"

The concept of *intention* is central to humans!

Global Understanding via Transparency

Definition (Levels of transparency)

Levels of *transparency* of a model:

- simulatable = understandable as a whole
- decomposable into simulatable parts
- algorithmically transparent = mathematical understanding

When is full transparency needed?

EU AI Act

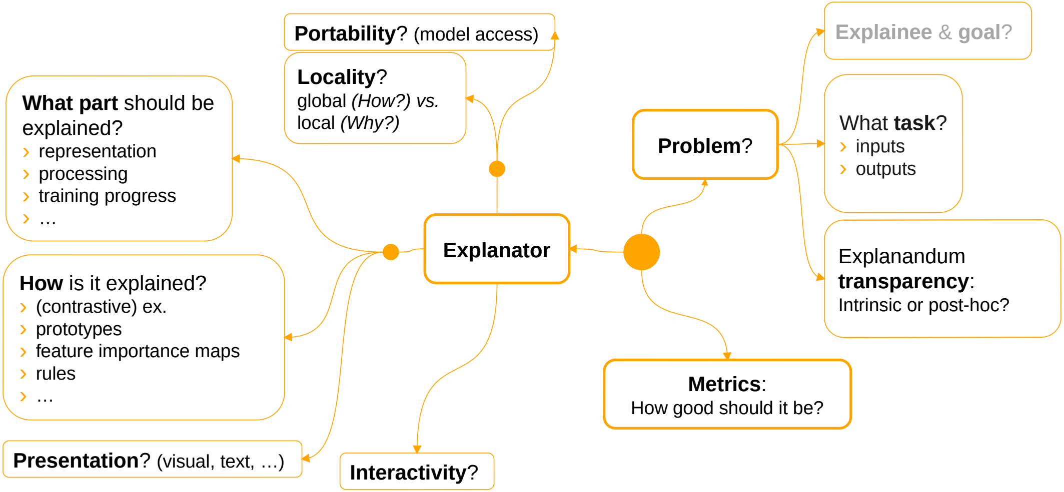
Preamble (72)

*[…] **transparency should be required for high-risk AI systems** […]. High-risk AI systems should be designed in a manner to enable deployers to **understand** how the AI system works, […]*

Article 13

*1. **High-risk AI systems** shall be designed and developed in such a way as to ensure that their operation is **sufficiently transparent** to enable deployers **to interpret** a system's output and **use it appropriately**. […]*

A Taxonomy of XAI Methods

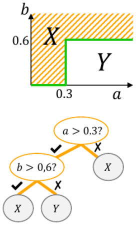


Outline

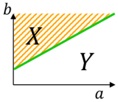
Inherently Interpretable Models

How does the model work?

- Decision **Trees**



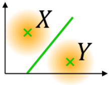
- **Linear** Models, Generalized **Additive** Models

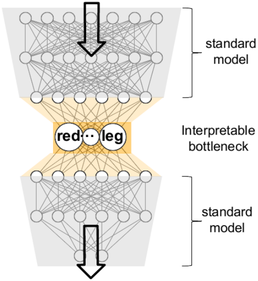


Linear: $f(x) = \alpha a + \beta b$

GAM: $f(x) = g^{-1}(f_a(a) + f_b(b))$

- **Clustering**

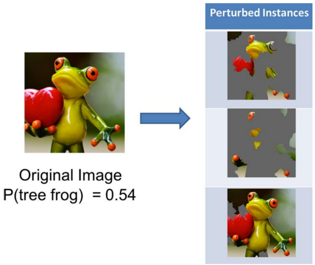




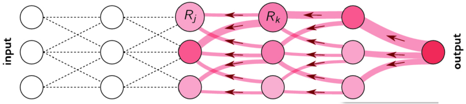
Feature Importance Methods and Contrastive Explanations

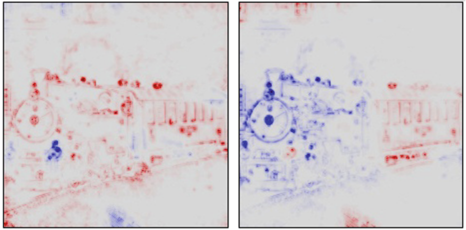
What features were important for the model's output?

Black-box, e.g., **perturbation**-based



White-box, e.g., **backpropagation**-based





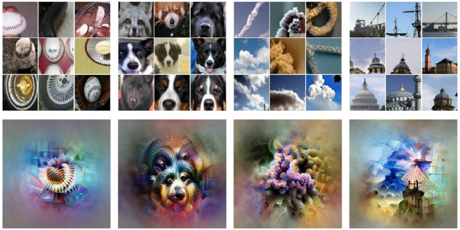
More advanced: *Why output y instead of y'?*

Concept-based Explainability for Explaining Representations

What does the output of a network unit/part (e.g., neuron, channel) encode?

Examples
activating unit strongly

DeepDream
Prototypes
= starting image
optimized to activate
unit strongly

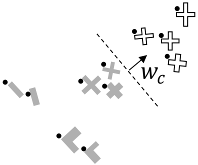


Baseball—or stripes? *mixed4a, Unit 6*
Animal faces—or snouts? *mixed4a, Unit 240*
Clouds—or fluffiness? *mixed4a, Unit 453*
Buildings—or sky? *mixed4a, Unit 492*

Goal: association

semantic concepts,
e.g., `isHead`

$\longleftrightarrow$



concept activation vectors w_c
(CAVs) in DNN latent space

Explainable Surrogates

What reasoning was learned?

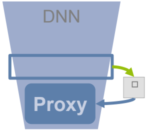
Idea: Approximate DNN (parts) by an interpretable model.
Examples:

- (Locally) **linear**,
- **Decision tree or rules**,
- Dependency / flow **graphs**,

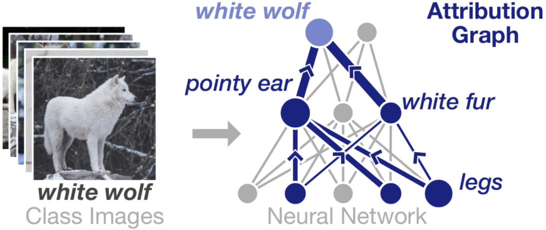
ResNeXt [12] classifying

face / no face

```
face(F) :- contains(F, A), isa(A, nose),
           contains(F, B), isa(B, mouth), top_of(A, B),
           contains(F, C), top_of(C, A)
```



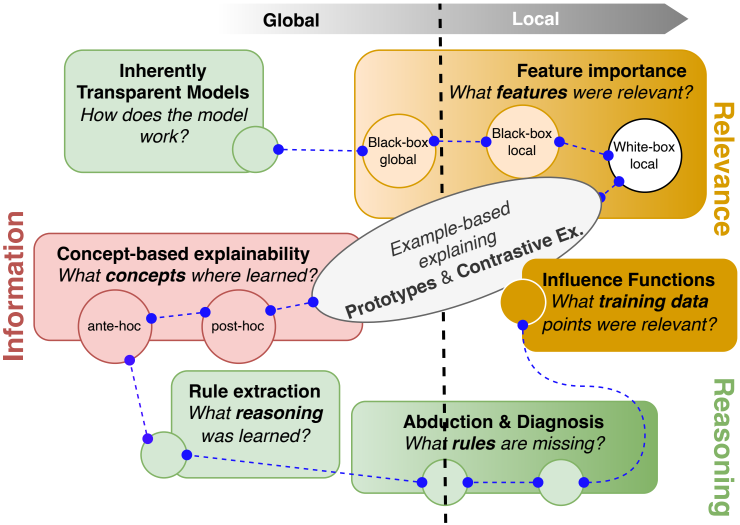
⋱



Other topics

- **Abduction and diagnosis:** *What rules are missing from the model?*
- **Explaining training data influence:** *What training data points were most influential?*
- **Applications:** Explainable Robotics
- **Quality of explanations**

Lecture Overview



Outline

What can go wrong with explanations

Consider a local explanation of the form (*deductive-nomological model* [4]):

$$\textit{Explanans}\begin{cases}\text{conditions } C \text{ fulfilled,}\\ \texttt{AND}\ \text{ general laws } L \text{ apply,}\end{cases}$$

switch turned off

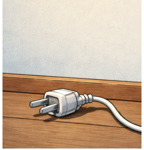
AND (switch off $\Rightarrow$ dark)

$$\Downarrow$$

$$\textit{Explanandum}\left\{\texttt{THUS}\quad \text{we observe } E.\right.$$

THUS dark.

Relevance!



Causality: *"Why is the switch off?"*

"Because it is dark, and if it is dark then the switch is (typically) off."

What makes “good” explanations?

Global explanations: (=“How does it work?”)

- Prefer transparent models where possible [10],
- mind the **explainability-accuracy tradeoff**

Local explanations (=“Why this decision?”):

- Use **contrastive** explanations (=“Why this decision *instead* of *another*?”) [8]
  E.g., “Why did I take the tram on Monday morning?”
  “… rather than walking?”
  “… rather than staying at home?”
  “… rather than riding the bicycle?””

What makes "good" explanations?

As of Miller, "Explanation in Artificial Intelligence" [8] (psychological perspective):

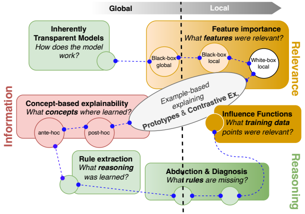
Mind the goal (**generality**): Enable the explainee to anticipate future states!

- Stay **faithful**, mind relevance!
- Prefer **causal links** over probabilities
- Maintain **transparency** by minimality/selectiveness!
- Cite the **abnormal** rather than normal causes.
- **Explainees' background** matters!
- **Context** matters for explanation selection! Stay **minimal**! *e.g., why is the explanation needed?*

Outline

Summary

- **Explanations** are intrisically cognitive:
  - for humans: causes, causal history, reason, and enabling factors of (un)intentional behavior (Malle [7])
  - in XAI: "help to understand aspects of a model" (learned information, influence, or system's reasoning)
- Many different methods available (very active field of research :-)), e.g.,
  - post-hoc vs. ante-hoc
  - local vs. global
  - model-agnostic vs. -specific
- **Good explanations** are in particular contrastive, faithful, causal, minimal, and respects explainee's background



Further Reading

- Schwalbe et al., “A Comprehensive Taxonomy for Explainable Artificial Intelligence” [11] (find hints for good introductory surveys here in the beginning)
- Christoph Molnar, *Interpretable Machine Learning* [2], Chap. 2
- Another classical philosophical model of explanation by Aristoteles: `https://en.wikipedia.org/wiki/Four_causes`

References I

[1] Sebastian Bruckert, Bettina Finzel, and Ute Schmid. "The Next Generation of Medical Decision Support: A Roadmap Toward Transparent Expert Companions". In: *Frontiers in Artificial Intelligence* 3 (Sept. 2020). ISSN: 2624-8212. DOI: 10.3389/frai.2020.507973. (Visited on 04/12/2026).

[2] Christoph Molnar. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd ed. 2025.

[3] C. A. E. Goodhart. "Problems of Monetary Management: The UK Experience". In: *Monetary Theory and Practice: The UK Experience*. Ed. by C. A. E. Goodhart. London: Macmillan Education UK, 1984, pp. 91–121. ISBN: 978-1-349-17295-5. DOI: 10.1007/978-1-349-17295-5_4. (Visited on 04/12/2026).

[4] Carl G. Hempel and Paul Oppenheim. "Studies in the Logic of Explanation". In: *Philosophy of Science* 15.2 (1948), pp. 135–175. DOI: 10.1086/286983.

[5] Been Kim, Rajiv Khanna, and Oluwasanmi O. Koyejo. "Examples Are Not Enough, Learn to Criticize! Criticism for Interpretability". In: *Advances in Neural Information Processing Systems* 29 (2016). (Visited on 04/12/2026).

[6] Carmen Lacave and Francisco Dez. "A Review of Explanation Methods for Bayesian Networks". In: *The Knowledge Engineering Review* 17 (May 2001). DOI: 10.1017/S026988890200019X.

[7] Bertram F. Malle. "How People Explain Behavior: A New Theoretical Framework". In: *Personality and Social Psychology Review* 3.1 (Feb. 1999), pp. 23–48. ISSN: 1088-8683. DOI: 10.1207/s15327957pspr0301_2. (Visited on 04/15/2026).

[8] Tim Miller. "Explanation in Artificial Intelligence: Insights from the Social Sciences". In: *Artificial Intelligence* 267 (Feb. 2019), pp. 1–38. ISSN: 0004-3702. DOI: 10.1016/j.artint.2018.07.007. (Visited on 10/11/2021).

[9] Milda Pocevičiūtė, Gabriel Eilertsen, and Claes Lundström. "Survey of XAI in Digital Pathology". In: *Lecture Notes in Computer Science* 2020 (2020), pp. 56–88. DOI: 10.1007/978-3-030-50402-1_4. arXiv: 2008.06353. (Visited on 02/24/2021).

[10] Cynthia Rudin. "Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead". In: *Nature Machine Intelligence* 1.5 (May 2019), pp. 206–215. ISSN: 2522-5839. DOI: 10.1038/s42256-019-0048-x. (Visited on 02/25/2021).

References II

[11] Gesina Schwalbe and Bettina Finzel. "A Comprehensive Taxonomy for Explainable Artificial Intelligence: A Systematic Survey of Surveys on Methods and Concepts". In: *Data Mining and Knowledge Discovery* 38 (Jan. 2023), pp. 3043–3101. ISSN: 1573-756X. DOI: 10.1007/s10618-022-00867-8. arXiv: 2105.07190. (Visited on 01/09/2023).

[12] Saining Xie et al. "Aggregated Residual Transformations for Deep Neural Networks". In: *Proc. 2017 IEEE Conf. Comput. Vision and Pattern Recognition*. Honolulu, HI, USA: IEEE Computer Society, 2017, pp. 5987–5995. DOI: 10.1109/CVPR.2017.634. arXiv: 1611.05431.